

Md Muntasir Hossain

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PROFILE

Doctoral researcher in Electrical and Computer Engineering at Lamar University working on deep space communication, Delay-Tolerant Networking, and the predictive modeling and security of LEO satellite networks. Combines field experimentation and machine learning with local-first AI systems engineering, with two IEEE conference publications, federally funded research, and a \$35,000 AWS-backed commercialization award.

EDUCATION

Doctoral of Engineering in Electrical and Computer Engineering 2026 – Present
Lamar University, Beaumont, TX

Research Focus: Deep space communication Earth to MARS and Delay-Tolerant Networking for intermittently connected and resource-constrained network environments.

Master of Science in Computer Science 2023 – May 2026
Lamar University, Beaumont, TX

Relevant Coursework: Software Engineering, Analysis of Algorithms, Computer Networks, Big Data, Machine Learning, Advanced Operating Systems.

Thesis: Predictive Performance Modeling and Security Analysis of Starlink LEO Satellite Networks in Maritime Environments

Bachelor of Science in Computer Science and Engineering 2017 – 2021
American International University-Bangladesh (AIUB), Dhaka

Relevant Coursework: Data Structures, Algorithms, Cloud Computing, Data Mining, Cybersecurity, Data Communication.

Thesis: Rigorous Study on Computational Geometry and Turing Test on Graph Layout Algorithms

- Explored algorithmic complexity and geometric optimization in graph layout models, evaluating performance through formal Turing test methodology.

PROFESSIONAL EXPERIENCE

Research Assistant Oct 2024 – Present
Center for Data AI and Cybersecurity (CDAC), Lamar University, Beaumont, TX

- Conducting research on performance evaluation and adaptive optimization of Starlink LEO satellite networks in maritime environments, under DOE and CAPM federal grants in collaboration with the Port of Beaumont.
- Extending this work into doctoral research on deep space communication and Delay-Tolerant Networking, targeting intermittent connectivity and long propagation delays in resource-constrained network environments.
- Investigating cybersecurity risks in intelligent maritime systems, including signal manipulation, spoofing, and data integrity threats.
- Delivered data-driven insights adopted by project stakeholders to strengthen connectivity and operational security for port activities.

Graduate Teaching Assistant Sep 2025 – Present
Lamar University, Beaumont, TX

- Assisting in undergraduate courses: Computer Networks (COSC 5328), Multimedia Processing (COSC 4330), and Introduction to Computer Architecture (COSC 4310).
- Supporting 50+ students per course through lab instruction, grading, and one-on-one academic guidance.
- Demonstrating tools such as Wireshark, NS-3, and Python-based network analysis to bridge theoretical and practical learning.

IT Student Assistant

Feb 2024 – Sep 2024

Lamar University, Beaumont, TX

- Delivered technical support to 500+ students, faculty, and staff, resolving hardware and software issues with a 95% first-contact success rate.
- Managed inventory of 1,000+ IT assets and optimized support databases, reducing troubleshooting time by 30%.
- Installed, configured, and maintained 200+ workstations, ensuring system reliability and high availability.

Web Developer

Sep 2021 – Jan 2023

RPSI Limited, Tech Team, Dhaka, Bangladesh

- Designed and deployed 3+ e-commerce and NGO websites using JavaScript, React, PHP, and Laravel, reducing deployment time by 20%.
- Contributed to a logistics management system that streamlined order tracking and reduced manual processing time by 30%.
- Collaborated with a 4-person development team, improving coordination and reducing delivery delays by 25%.

PROJECTS AND SOFTWARE

LEO SatLink Observatory — Desktop Application and Web Dashboard

2025 – Present

Python, Tkinter, Scikit-learn, Pandas, NumPy, React, Next.js, TypeScript, Recharts

- Developed a self-contained desktop application and companion React/Next.js decision-support dashboard for real-world Starlink performance monitoring and forecasting in maritime environments, serving as a core deliverable of the M.S. thesis.
- Implemented a continuous background polling engine (30-second cycle) over a rolling 30-minute inference window, with automatic re-evaluation after each poll cycle.
- Engineered a dual-task Random Forest forecasting engine predicting downlink throughput and latency 15 minutes ahead simultaneously, with health scoring, safe-duration estimation, and early warning boundary detection.
- Applied 1-minute resampling of the published empirical 1,000-row coastal dataset to maintain model compatibility, preserving dataset integrity across both published conference papers.

AI Agentic Workflow Orchestration for System Development

2026 – Present

Python, LangGraph, LangChain, Ollama, Model Context Protocol, FastMCP, ChromaDB, tree-sitter, Langfuse

- Built a four-agent LangGraph state machine (Planner, Developer, Tester, Deploy) coordinating the full software development lifecycle entirely on local NVIDIA hardware, with on-device embedding models keeping proprietary codebases off external services.
- Engineered surgical-mcp-rag, a write-capable Model Context Protocol server delivering AST-aware, function-level code retrieval through tree-sitter and ChromaDB, with end-to-end execution tracing via Langfuse.

ICS-ThreatLens — Local-First ICS Threat Intelligence CLI

2026

Python, MITRE ATT&CK for ICS, YAML

- Developed a command-line tool that parses public ICS/SCADA security advisories, maps them to MITRE ATT&CK for ICS techniques, and cross-references them against a private local asset inventory to generate prioritized, operator-specific risk reports.
- Enforced a strict trust boundary in which only public advisory text reaches the AI parser while asset inventory data never leaves the host, including an offline fixture mode for air-gapped operational technology environments.

Token Anatomy — Local LLM Usage Analytics Dashboard

2026

Python (zero-dependency), SQLite, HTML/CSS/JavaScript

- Built a zero-dependency, self-hosted dashboard that parses local LLM coding-session logs into token, cost, and cache-hit analytics across a 60-day window, served over localhost with 60-second auto-refresh and no data leaving the machine.

- Implemented hourly usage heatmaps, session topic clustering, per-project cost attribution, and an advisory engine producing seven plain-language optimization diagnoses, with configurable pricing for AWS Bedrock and direct API rates.

Jetson-Postboot — Edge AI Provisioning Tool

2026

Python, Bash, NVIDIA JetPack 6.x, CUDA, Ollama

- Created a single-command diagnostic and remediation tool automating post-boot configuration of NVIDIA Jetson Orin Nano developer kits, covering SD-to-SSD storage reclamation, swap tuning for model inference, CUDA and Ollama health verification, and stale boot-order detection.
- Designed for non-invasive operation: read-only on first run, explicit confirmation and dry-run mode before any change, and fully reversible modifications isolated to the project directory, with PASS/WARN/ACTION reporting in plain language.

PUBLICATIONS

A Dual-Task Prediction Model for Starlink Maritime Performance

Feb 2026

5th IEEE International Conference on AI in Cybersecurity (ICAIC 2026), University of Houston, TX

Richard Li, Md Muntasir Hossain, Xingya Liu, Ruhai Wang

- Developed a short-term dual-task prediction model predicting downlink throughput and latency simultaneously using a shared Random Forest backbone, outperforming single-task baselines. Trained in real-world coastal experimental data.

Predictive Model for Starlink Maritime Performance Using Multi-Horizon RandomForest

2026

IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC 2026)

Md Muntasir Hossain, Xingya Liu, Helen H. Lou, Ruhai Wang, Kazi Fazlee Rabbi

- Proposed a data-driven multi-horizon Random Forest regression model for downlink throughput forecasting, trained on real coastal measurement data incorporating network-layer indicators and environmental variables. Reduced mean absolute error by approximately 31% over a persistence baseline at a 15-minute horizon.

AWARDS AND GRANTS

2nd Place AWS AI Pitch Competition

April 2026

Small Business Development Center at Lamar University x Amazon Web Services

- Awarded \$35,000 prize for SatLink AI, a predictive connectivity intelligence platform for maritime and port operations, recognized for bridging academic research with entrepreneurial commercialization for Gulf Coast port operators.

Dr. Anwarul Abedin Scholarship Grant

2017 – 2021

American International University-Bangladesh

Awarded for outstanding academic performance and consistent excellence throughout undergraduate studies.

Dean's List Award

2019

American International University-Bangladesh

Recognized for academic excellence and ranking among the top students in the Department of Computer Science.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, PHP, SQL, Bash

Frameworks & Libraries: React, Next.js, Scikit-learn, NumPy, Pandas, Matplotlib, Tkinter, LangChain, LangGraph, n8n workflow. Higgsfield MCP, Novamira MCP, Autodesk MCP.

Tools & Platforms: NS-3, Wireshark, Ollama, Model Context Protocol (MCP), ChromaDB, Langfuse, tree-sitter, NVIDIA Jetson / JetPack, CUDA, GitHub, Overleaf, Jira, VirtualBox, Netwox, OmniRoute, Vllm, Pix4DMapper, ArcGIS Pro 3.6.

Cybersecurity: Network Spoofing, Sniffing, Session Hijacking, DoS/DDoS, ICS/SCADA Security, MITRE ATT&CK for ICS (CISA 201 Certified),

Databases: MySQL, MongoDB, SQLite